

Girish Yandigeri

PROGRAM – 7

Commands

`minikube start`

Starts the Minikube Kubernetes cluster.

`minikube status`

Shows the current status of the Minikube cluster.

`minikube delete`

`minikube start --driver=docker`

Deletes any old cluster and starts Minikube using the Docker driver.

`kubectl apply -f nginx-deployment.yaml`

Creates the NGINX deployment from the YAML file.

`kubectl get deployments`

`kubectl get pods`

Displays the running deployments and pods.

`kubectl expose deployment nginx-deployment --type=NodePort --port=80`

Creates a service and exposes the deployment on port 80.

`kubectl get svc`

Shows the list of services running in the cluster.

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ nano nginx-deployment.yaml
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  0/2     2            0           6s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ kubectl get pods
NAME          READY   STATUS             RESTARTS   AGE
nginx-deployment-6f9664446b-cxfs6  0/1     ContainerCreating   0          13s
nginx-deployment-6f9664446b-dchkr  0/1     ContainerCreating   0          13s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ kubectl get services
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes    ClusterIP     10.96.0.1    <none>         443/TCP          7d9h
nginx-deployment  NodePort      10.110.118.78 <none>        80:30668/TCP     5s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ minikube service nginx-deployment

```

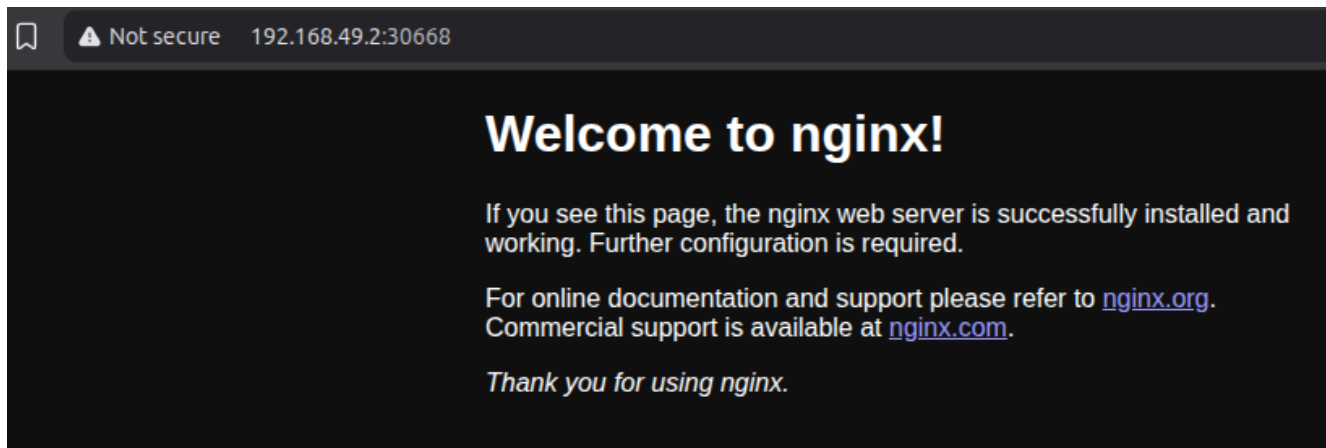
NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30668

```

🌐 Opening service default/nginx-deployment in default browser...
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0:~/devops/nor...$ Opening in existing browser session.
```

`minikube service nginx-deployment`

Opens the NGINX service in the default browser.



```
kubectl delete pods --all
```

Deletes all running pods.

```
kubectl delete svc --all
```

Deletes all created services.

```
kubectl delete deployments --all
```

Deletes all deployments.

YAML File: nginx-deployment.yaml

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: nginx-deployment
```

```
spec:
```

```
  replicas: 2
```

```
  selector:
```

```
    matchLabels:
```

```
      app: nginx
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: nginx
```

```
    spec:
```

```
      containers:
```

```
      - name: nginx
```

```
        image: nginx:latest
```

```
        ports:
```

```
        - containerPort: 80
```